

supervisor	
signature	

* You should write your answer in English. Otherwise, small amount of penalty may be applied.

(5) If you do not make any constructor in class, the compiler would generate ().

5. (29points) The class for "complex number" (복소수) can be often defined using operator overloading. Write your implementation of member functions (1) **operator-** and (2) **operator<<**.

In (1) and (2), you may use standard library in your code. Your code should be grammatically and logically correct.

```
-----
class Complex {
private:
    double re, im;
public :
    Complex( double r, double i ) : re(r), im(i) {}
    friend Complex operator-(Complex&, Complex&);
    // other member functions for operator overloading are defined.
    friend ostream& operator<<(ostream &o, Complex &x);
    // prints x in "a+bi" format. for example, if re=3.2 and im=2.3, then this fuction should print 3.2+2.3i
    // if re=3.2 and im=-2.3, then this function should print 3.2-2.3i
    // if re=3.2 and im=0, then this function should print 3.2
};
-----
```

(1) Write C++ code for the **operator-** function

Complex operator-(Complex& x, Complex& y)

```
{
    
}
```

(2) Write C++ code for the **operator<<** function.

ostream& operator<<(ostream &o, Complex& x)

```
{
    
}
```

6. (12 points) Consider following C++ code and its output for area calculation. Fill out empty boxes with appropriate C++ code.

```
1: #include <iostream>
2: using namespace std;
3: class Rectangle {
4:     int width, height;
5: public :
6:     Rectangle(int x,int y) : width(x),height(y) {}
7:     int area() { return width*height; }
8: };
9: int main() {
10:     Rectangle obj(3,4);
11:     Rectangle *foo, *bar, *baz;
12:     foo = &obj;
13:     bar = new Rectangle (5,6);
14:     baz = new Rectangle[2] { {2,5}, {3,6} };
15:     cout << "Area Calculation Output\n";
16:     cout << "obj:" <<  << ", ";
17:     cout << "foo:" <<  << ", ";
18:     cout << "bar:" <<  << ", ";
19:     cout << "baz[0]:<<  << "\n";
20:     delete bar; delete[] baz; return 0;
21: }
```

Area Calculation Output

obj:12, foo:12, bar:30, baz[0]:10

7. (9 points) Consider following C++ code and its output showing the size of input file "example.bin". Fill out empty boxes with appropriate C++ code.

```
1: #include <iostream>
2: #include <fstream>
3: using namespace std;
4:
5: int main() {
6:     streampos begin, end;
7:     ifstream myfile("example.bin", ios::binary);
8:     begin =  ;
9:     myfile.  (0,ios::end);
10:    end=  ;
11:    myfile.close();
12:    cout << "size is: " << (end-begin) << " bytes\n";
13:    return 0;
14: }
```

Output: size is: 40 bytes